Optional: Deploy guestbook app from the OpenShift internal registry

Skills Network

Objectives

In this lab, you will:

- Build and deploy a simple guestbook application
- Use OpenShift image streams to roll out an update
- Deploy a multi-tier version of the guestbook application

Pre-requisite

You must have built and pushed the Guestbook application using the Docker commands as given in the Final Assignment.

Deploy guestbook app from the OpenShift internal registry

As discussed in the course, IBM Cloud Container Registry scans images for common vulnerabilities and exposures to ensure that images are secure. But OpenShift also provides an internal registry – recall the discussion of image streams and image stream tags. Using the internal registry has benefits too. For example, there is less latency when pulling images for deployments. What if we could use both—use IBM Cloud Container Registry to scan our images and then automatically import those images to the internal registry for lower latency?

Please continue with the below commands from the steps where you left off in the previous lab.

1. Create an image stream that points to your image in IBM Cloud Container Registry.

```
1. 1
1. oc tag us.icr.io/$MY_NAMESPACE/guestbook:v1 guestbook:v1 --reference-policy=local --scheduled
```

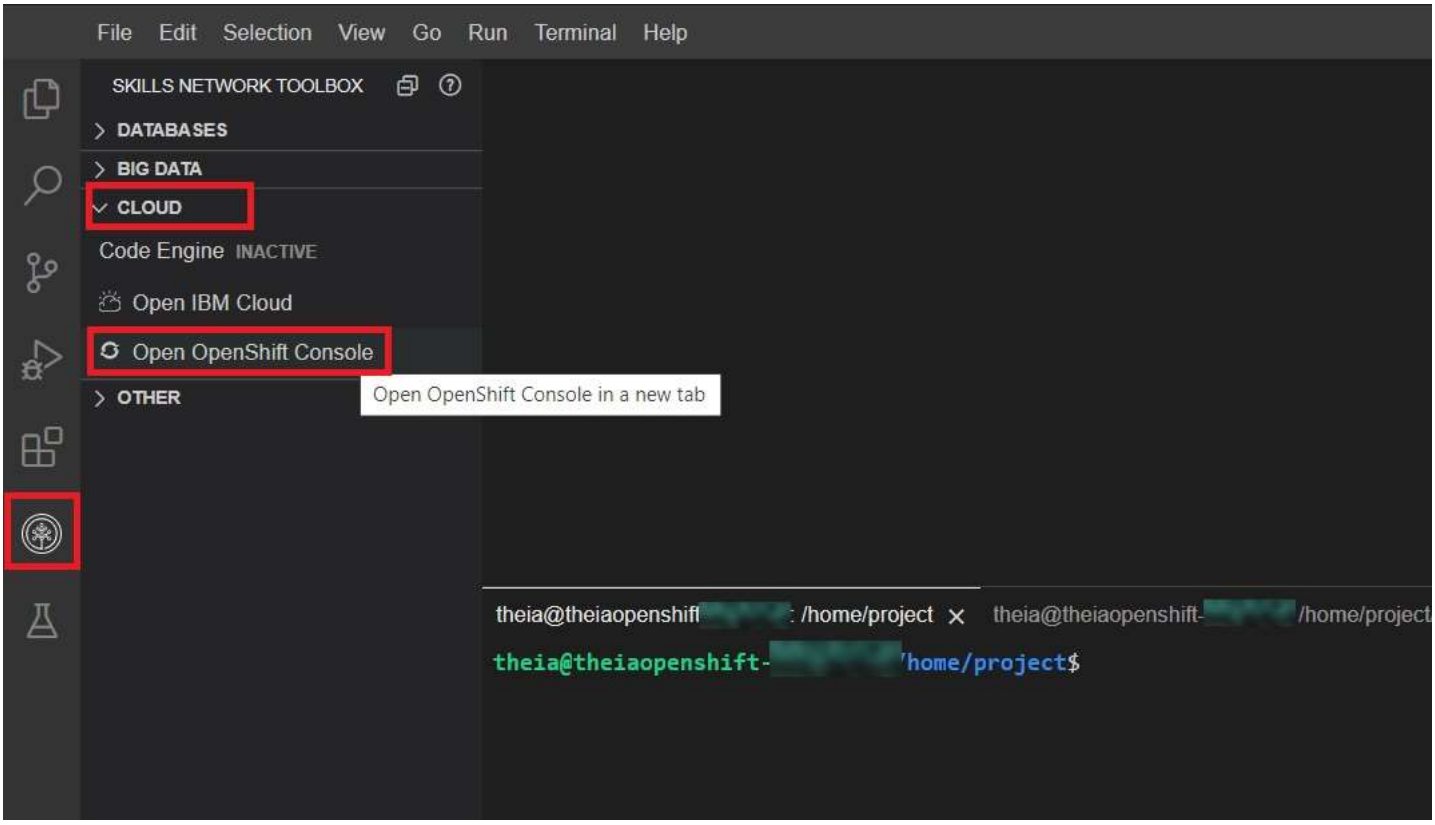
Copied!

With the `--reference-policy=local` option, a copy of the image from IBM Cloud Container Registry is imported into the local cache of the internal registry and made available to the cluster’s projects as an image stream. The `--schedule` option sets up periodic importing of the image from IBM Cloud Container Registry into the internal registry. The default frequency is 15 minutes.

```
theia@theiaopenshift-[redacted]:/home/project/guestbook/v1/guestbook$ oc tag us.icr.io/$MY_NAMESPACE/guestbook:v1 guestbook:v1 --re
ed
Tag guestbook:v1 set to import us.icr.io/sn-labs-[redacted]/guestbook:v1 periodically.
theia@theiaopenshift-[redacted]:/home/project/guestbook/v1/guestbook$
```

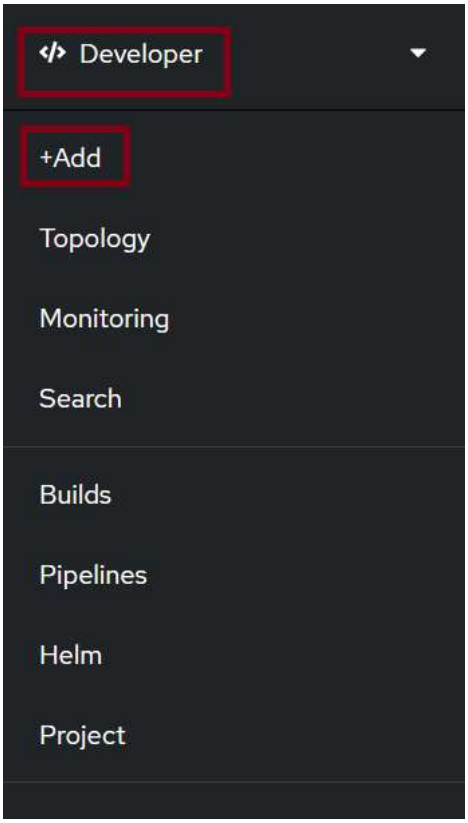
Now let’s head over to the OpenShift web console to deploy the guestbook app using this image stream.

2. Click on the `Skills Network Tool` and select `Cloud` which will open a drop-down. Click on `Open OpenShift console` which will open the Open Shift Web console in a new window.

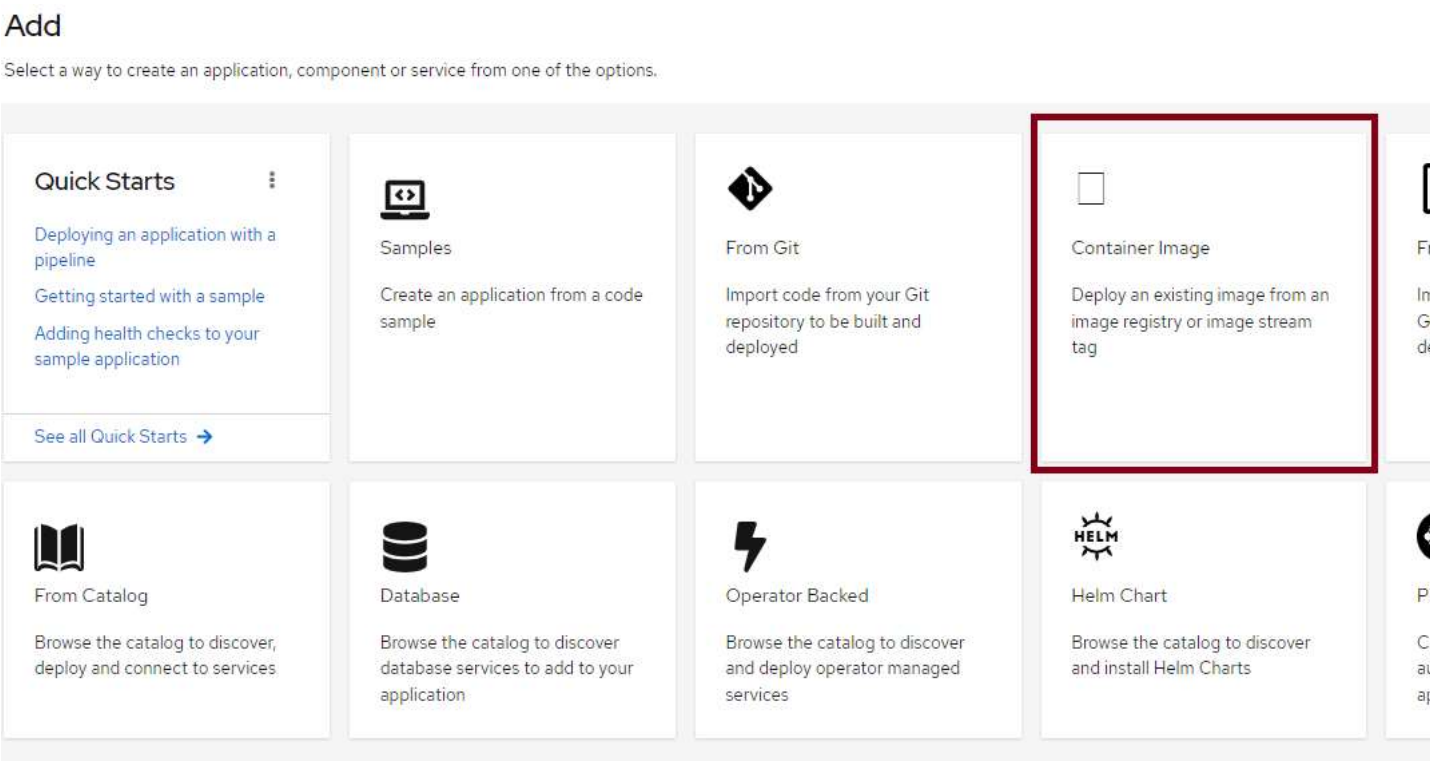


Note: Currently we are experiencing certain difficulties with the OpenShift console . If your screen takes time in loading, please close the OpenShift console browser tab & re-launch the same. It may take upto 10 mins to load the screen.

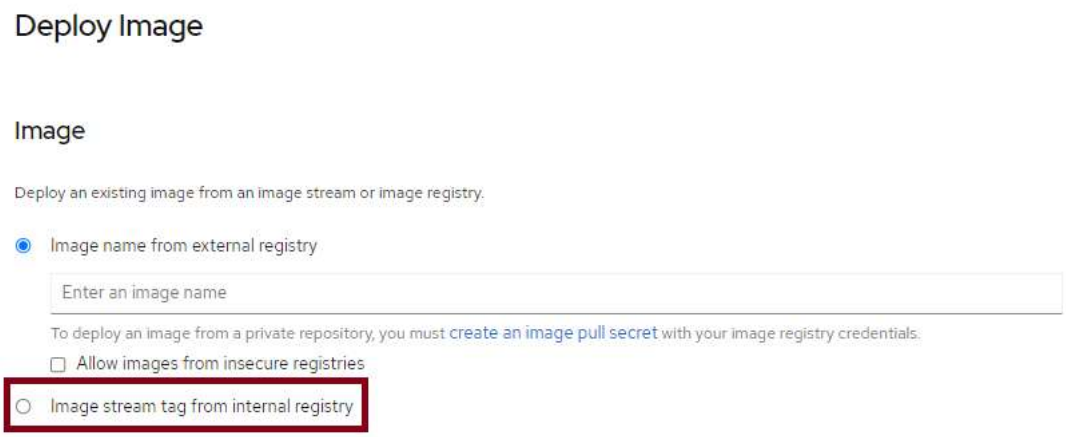
3. From the Developer perspective, click the **+Add** button to add a new application to this project.



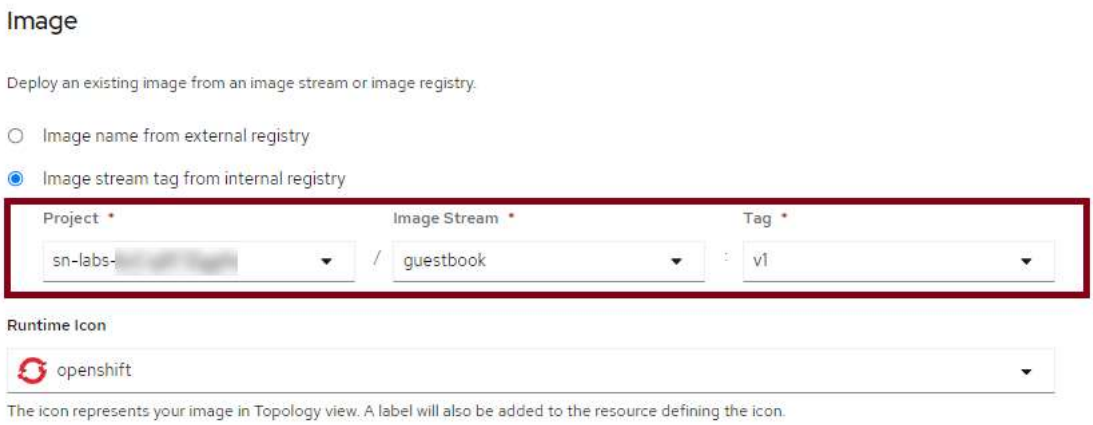
4. Click the **Container Image** option so that we can deploy the application using an image in the internal registry.



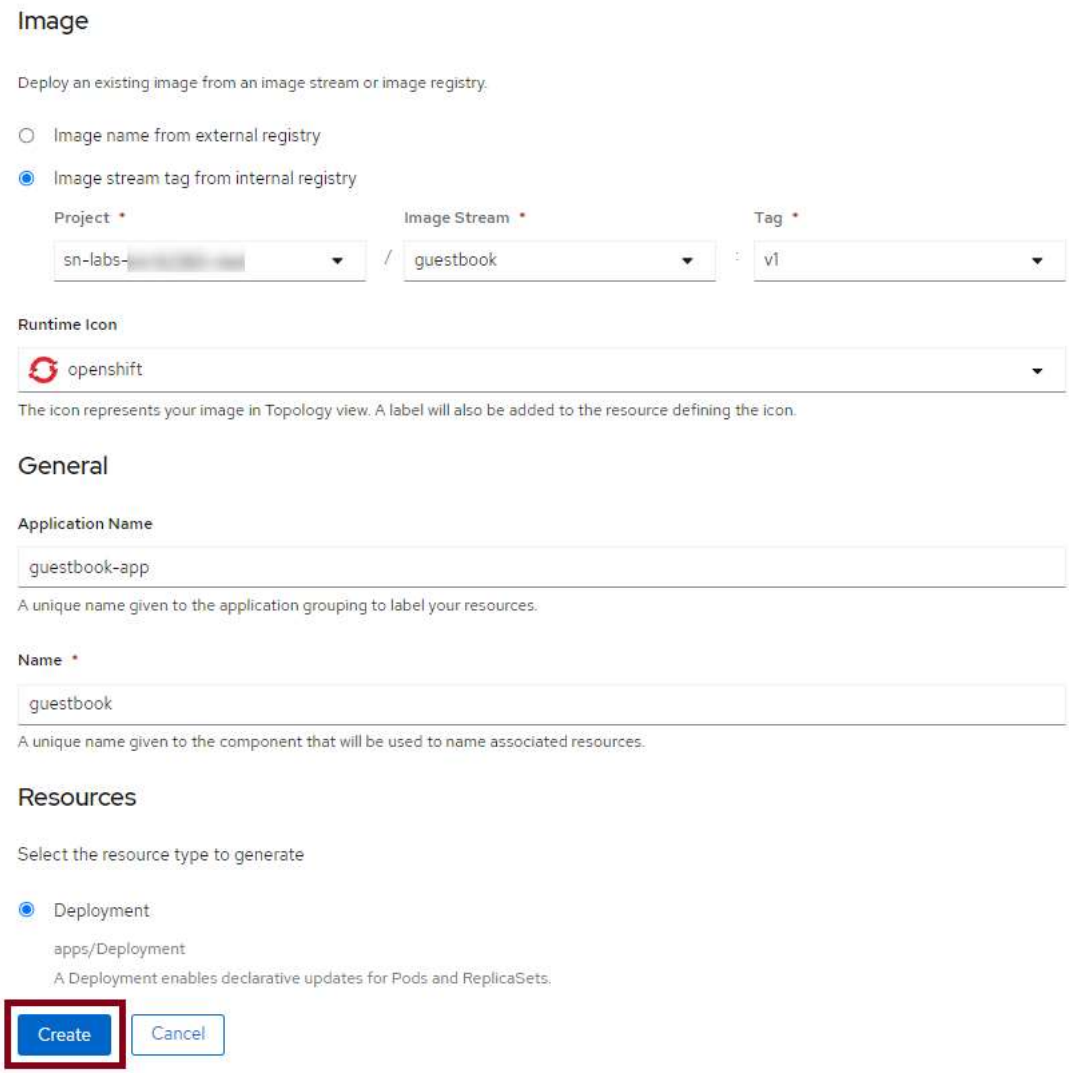
5. Under **Image**, switch to "**Image stream tag from internal registry** ".



6. Select your project, and the image stream and tag you just created (`guestbook` and `v1`, respectively). You should have only have one option for each of these fields anyway since you only have access to a single project and you only created one image stream and one image stream tag.

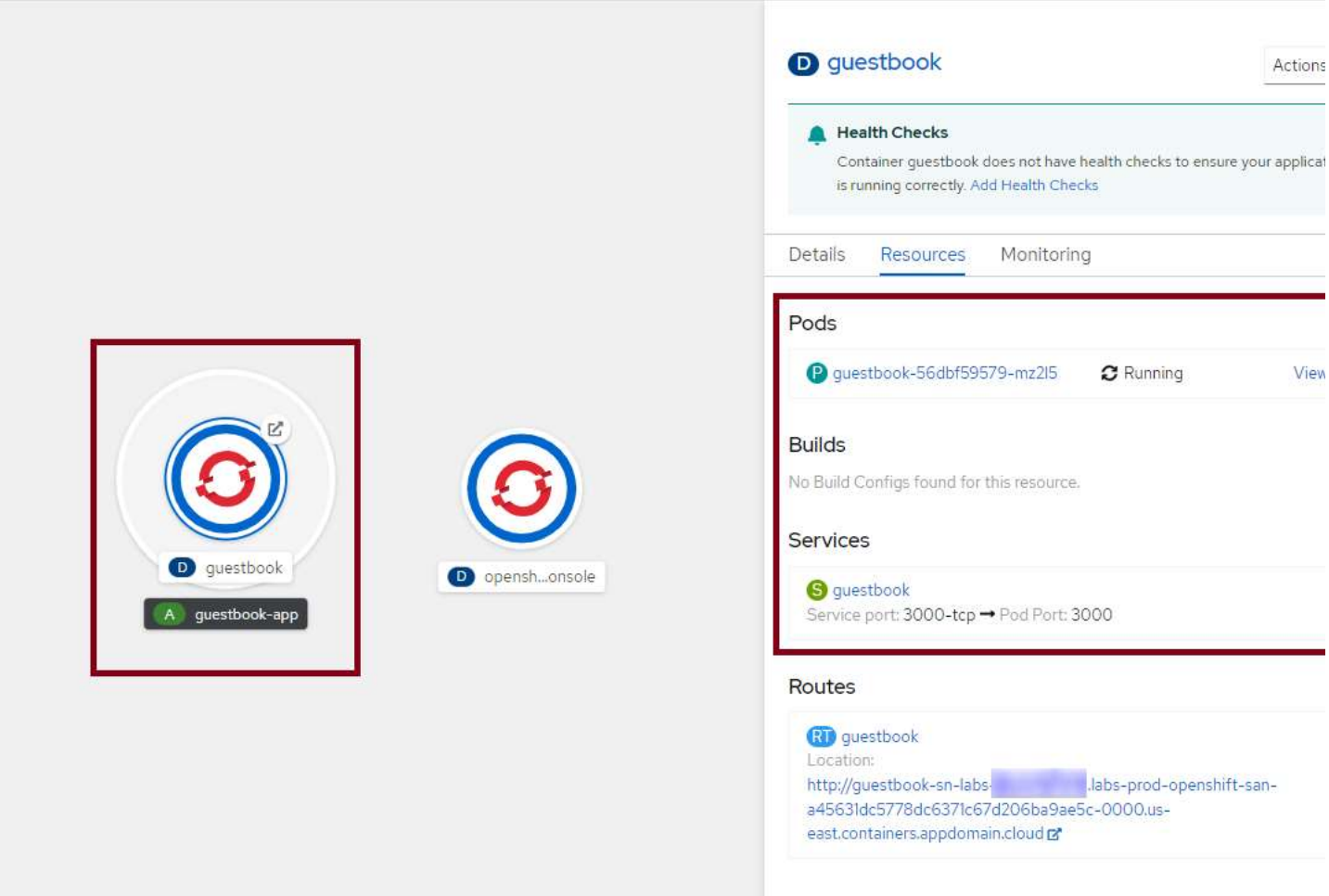


7. Keep all the default values and hit **Create** at the bottom. This will create the application and take you to the Topology view.



8. From the Topology view, click the `guestbook` Deployment. This should take you to the **Resources** tab for this Deployment, where you can see the Pod that is running the application as well as the Service and Route that expose it.

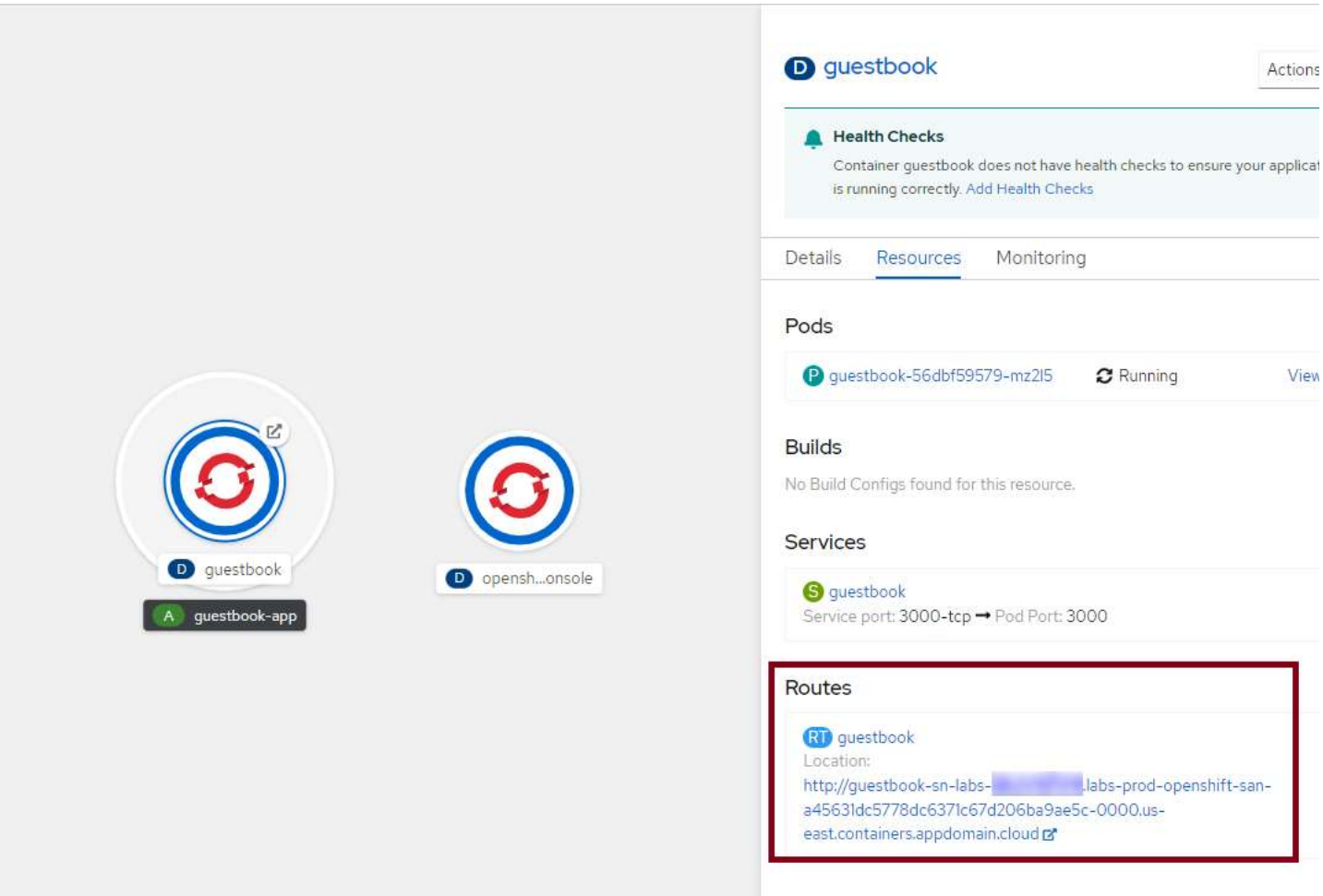
Kindly wait as the deployments in the Topology view may take time to get running.



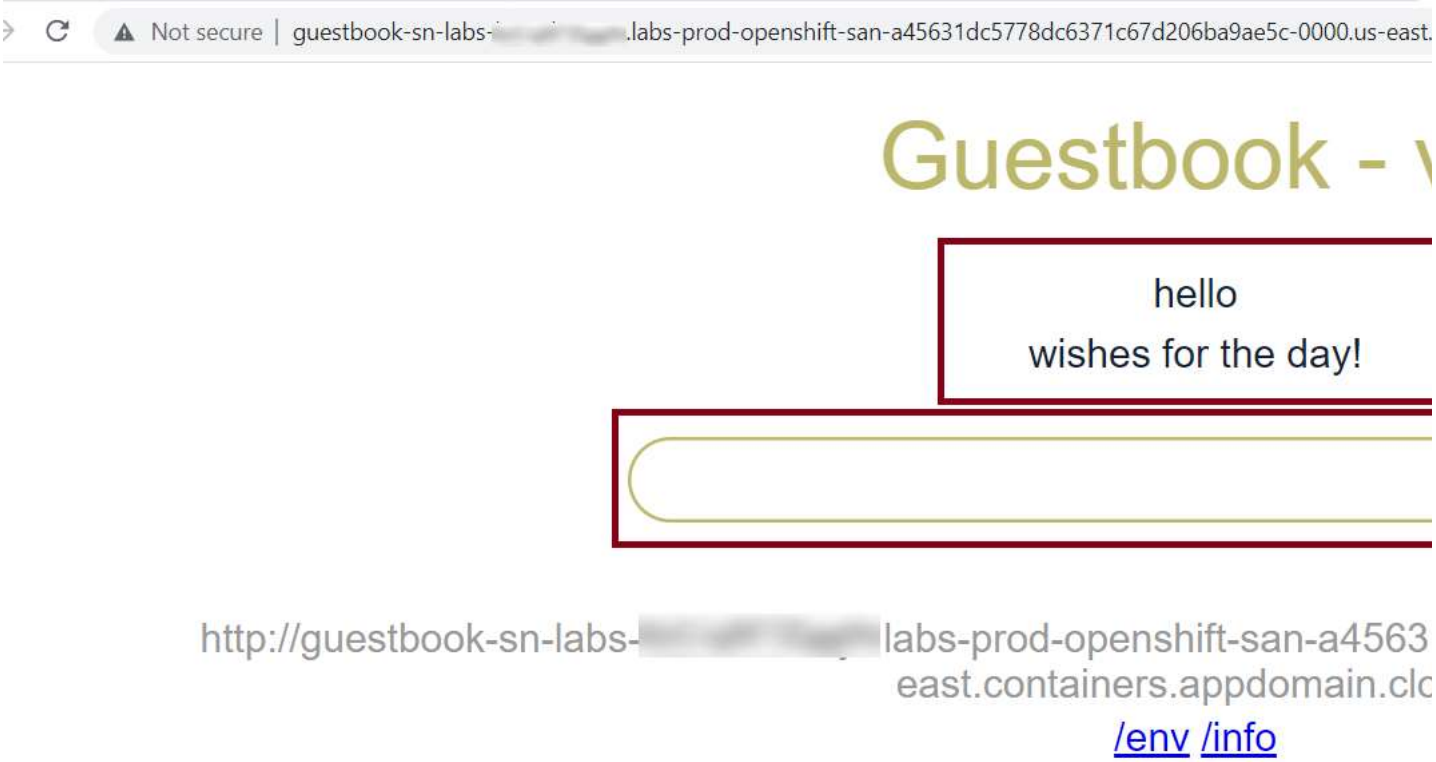
Note: Kindly do not delete the `opensh. console` **deployment in the Topography view as this is essential for the OpenShift console to function properly.**

9. Click the Route location (the link) to view the guestbook in action.

   Note: Please wait for status of the pod to change to **'Running'** before launching the app.



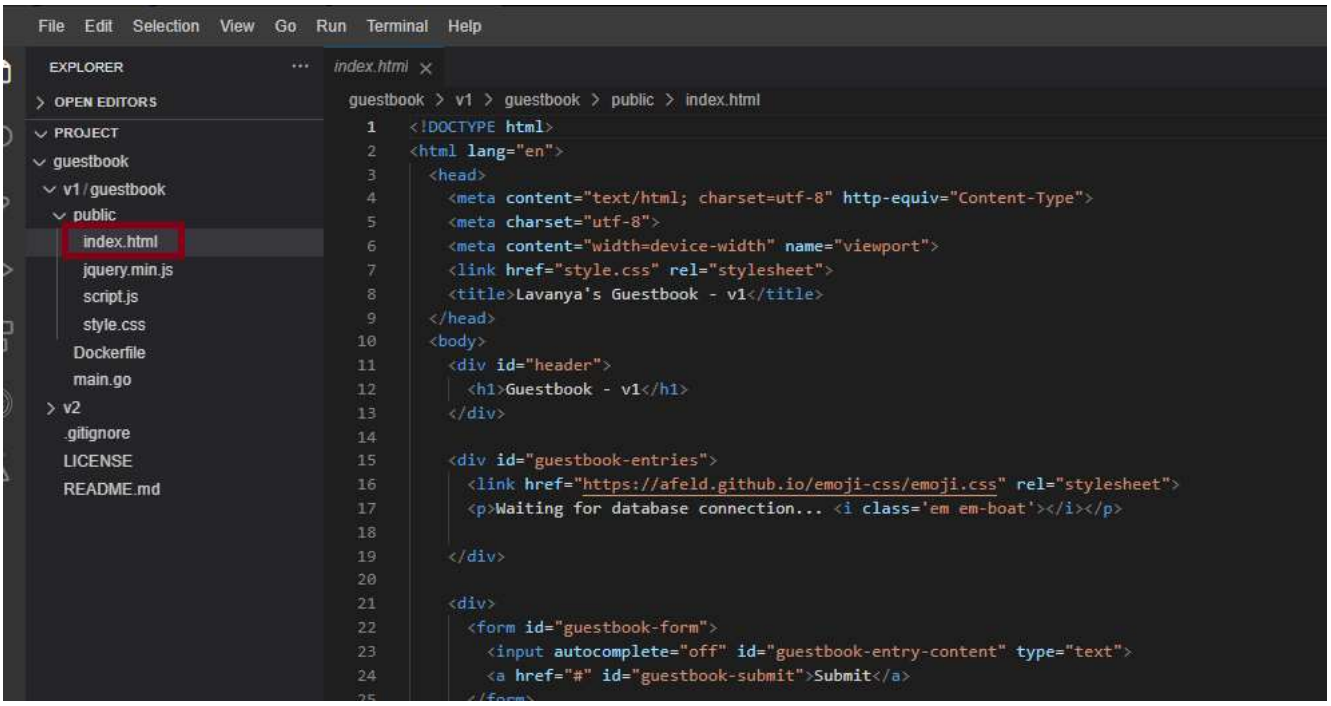
10. Try out the guestbook by putting in a few entries. You should see them appear above the input box after you hit **Submit**.



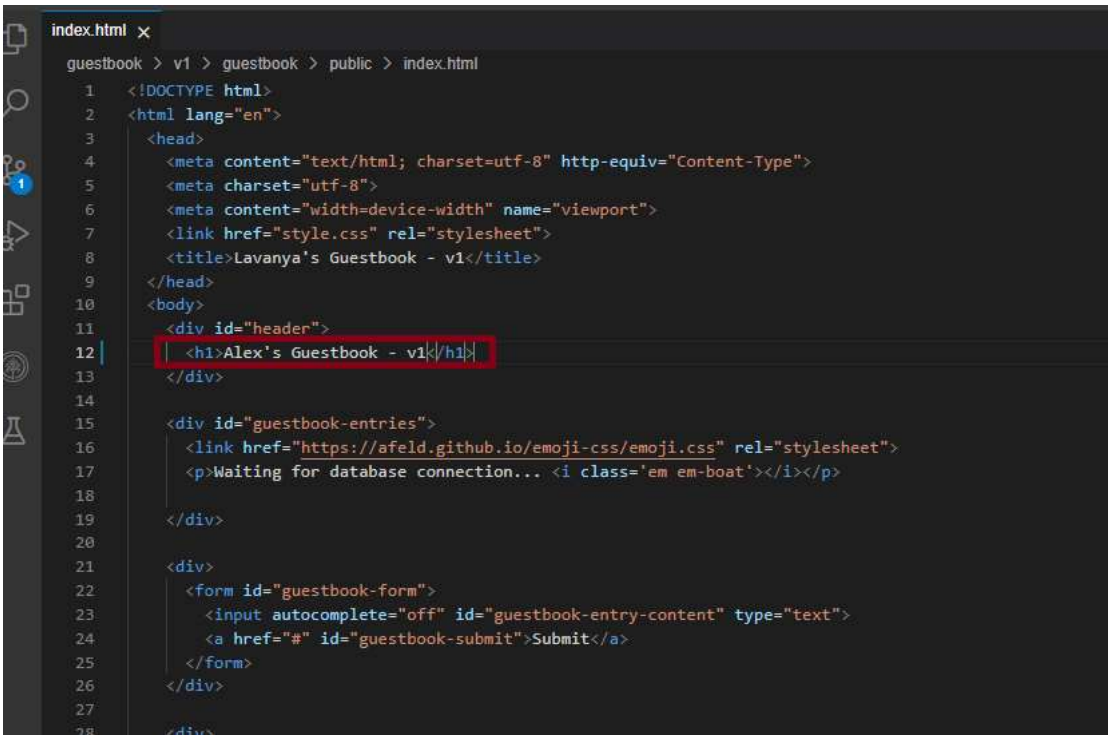
Update the guestbook

Let’s update the guestbook and see how OpenShift’s image streams can help us update our apps with ease.

1. Use the Explorer to edit `index.html` in the `public` directory. The path to this file is `guestbook/v1/guestbook/public/index.html`.



2. Let’s edit the title to be more specific. On line number 12, that says `<h1>Guestbook - v1</h1>`, change it to include your name. Something like `<h1>Alex’s Guestbook - v1</h1>`. Make sure to save the file when you’re done.



3. Build and push the app again using the same tag. This will overwrite the previous image.

```
1
docker build . -t us.icr.io/$MY_NAMESPACE/guestbook:v1 && docker push us.icr.io/$MY_NAMESPACE/guestbook:v1
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v1/guestbook$ docker build . -t us.icr.io/$MY_NAMESPACE/guestbook:v1 && doc
ook:v1
Sending build context to Docker daemon   98.3kB
Step 1/14 : FROM golang:1.15 as builder
 ---> 40349a2425ef
Step 2/14 : RUN go get github.com/codegangsta/negroni
 ---> Using cache
 ---> a7657fc96c64
Step 3/14 : RUN go get github.com/gorilla/mux github.com/xyproto/simpleredis
 ---> Using cache
 ---> 1f28b8fef54e
Step 4/14 : COPY main.go .
 ---> Using cache
 ---> 3458048c5c1e
Step 5/14 : RUN go build main.go
 ---> Using cache
 ---> 823973fe49e6
Step 6/14 : FROM ubuntu:18.04
 ---> f5cbed4244ba
Step 7/14 : COPY --from=builder /go//main /app/guestbook
 ---> Using cache
 ---> 0350722f466e
Step 8/14 : ADD public/index.html /app/public/index.html
 ---> a9644611258d
Step 9/14 : ADD public/script.js /app/public/script.js
 ---> 89215eb5fecd
Step 10/14 : ADD public/style.css /app/public/style.css
 ---> 7760ade3c7d3
Step 11/14 : ADD public/jquery.min.js /app/public/jquery.min.js
 ---> 97abdf76f88d
Step 12/14 : WORKDIR /app
 ---> Running in 1b78236b819b
Removing intermediate container 1b78236b819b
```

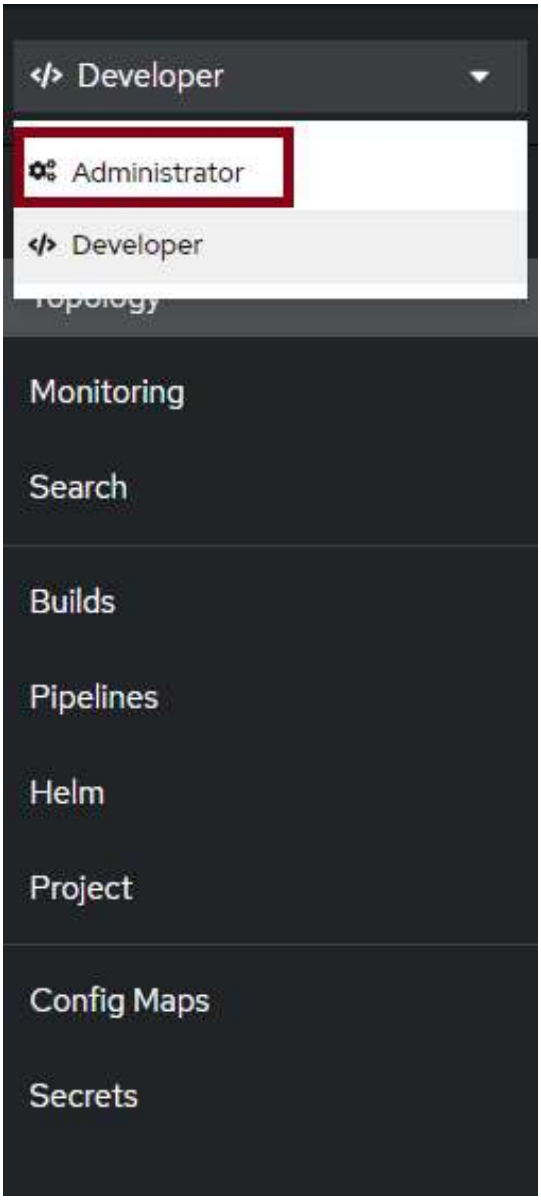
4. Recall the `--schedule` option we specified when we imported our image into the OpenShift internal registry. As a result, OpenShift will regularly import new images pushed to the specified tag. Since we pushed our newly built image to the same tag, OpenShift will import the updated image within about 15 minutes. If you don’t want to wait for OpenShift to automatically import the image, run the following command.

```
1. 1
1. oc import-image guestbook:v1 --from=us.icr.io/$MY_NAMESPACE/guestbook:v1 --confirm
```

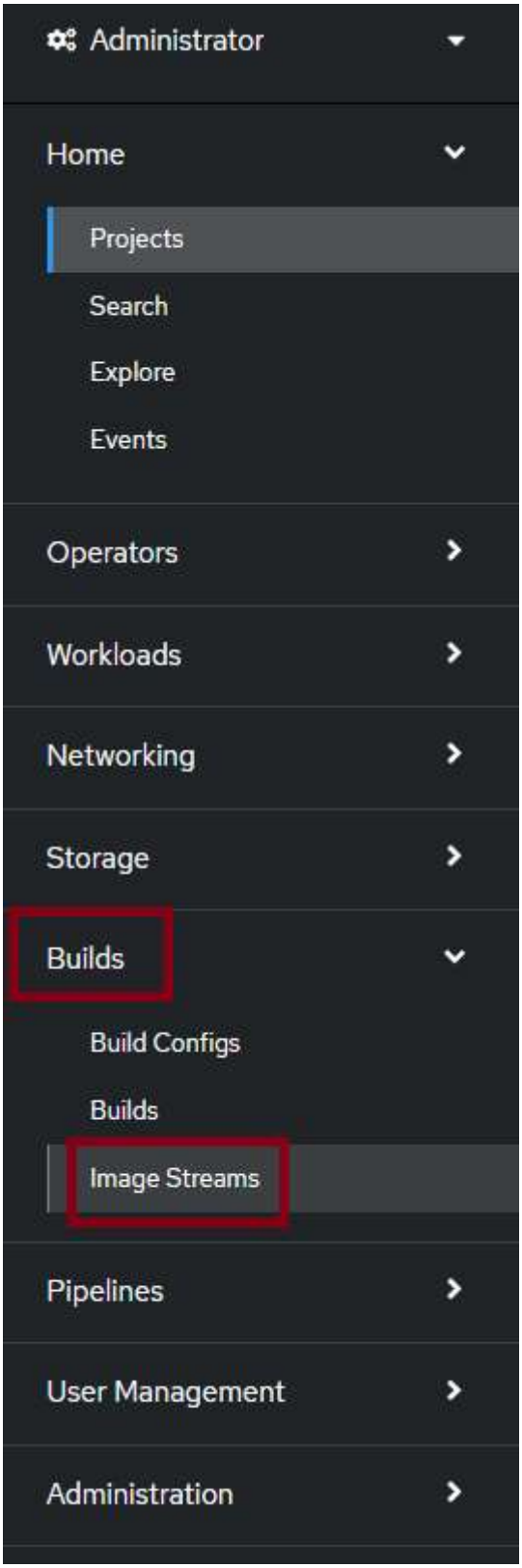
Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v1/guestbook$ oc import-image guestbook:v1 --from=us.icr.io/$MY_NAMESPACE/g
imagestream.image.openshift.io/guestbook imported

Name:                   guestbook
Namespace:              sn-labs-[redacted]
Created:                2 minutes ago
Labels:                 <none>
Annotations:            openshift.io/image.dockerRepositoryCheck=2022-04-11T08:28:50Z
Image Repository:       image-registry.openshift-image-registry.svc:5000/sn-labs-[redacted]guestbook
Image Lookup:           local=false
Unique Images:          2
Tags:                   1

v1
  updates automatically from registry us.icr.io/sn-labs-[redacted]/guestbook:v1
    prefer registry pullthrough when referencing this tag

  * us.icr.io/sn-labs-[redacted]/guestbook@sha256:f16401f8452ae414e5feafbb621ef20a0b9a12aafc800679465d809525ee454d
      Less than a second ago
    us.icr.io/sn-labs-[redacted]/guestbook@sha256:11ee56c6d46d80f0fda0790f50121bf3a4760240d74baef01a5c6bded6e94ef7
      2 minutes ago

Image Name:     guestbook:v1
Docker Image:   us.icr.io/sn-labs-[redacted]/guestbook@sha256:f16401f8452ae414e5feafbb621ef20a0b9a12aafc800679465d809525ee454d
Name:           sha256:f16401f8452ae414e5feafbb621ef20a0b9a12aafc800679465d809525ee454d
Created:        Less than a second ago
Annotations:    image.openshift.io/dockerLayersOrder=ascending
Image Size:     31.53MB in 6 layers
Layers:         26.71MB sha256:08a6abff89437fab99b52abbefed82ea907f12845c30eeb94f6b93c69be93166
                4.791MB sha256:1158f7aa125a353046e120906d415a3f0e7b2fb43fa61ffed49645f9eb423948
                650B    sha256:98e5b3b04689c7d80aebec108d7d88619d06fe3771382c6fa44d7273b74d5faa
                608B    sha256:e3277739aac4aaac2fbabf0e282914400e62c5e67da5cea7825eb5b0a519aca0
                545B    sha256:5a480e70e8fe40c001754b4143e4bb13a945b38e5adf7082a83ef70145d45b0d
                29.88kB sha256:5e702cd921b48b45d5f1ad2fa8d75390a12189f4c249acf995cc5185f3015a83
Image Created:  28 seconds ago
Author:         <none>
Arch:           amd64
```

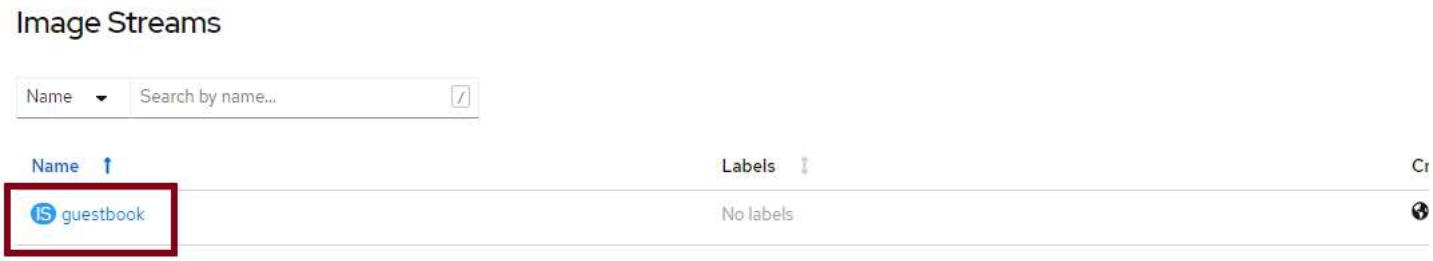
5. Switch to the **Administrator** perspective so that you can view image streams.



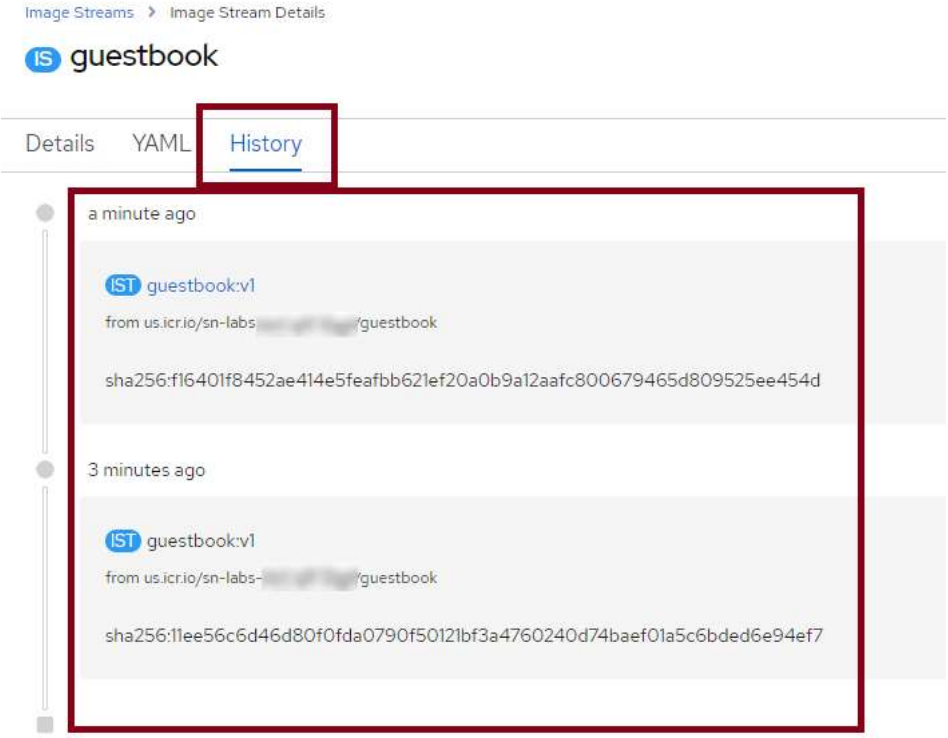
6. Click **Builds** > **Image Streams** in the navigation.



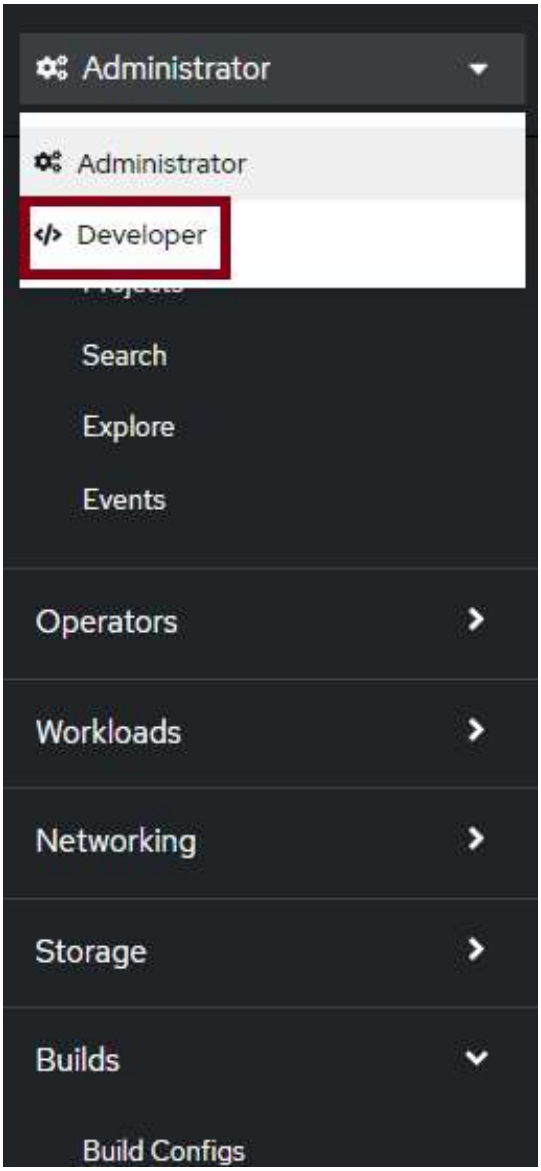
7. Click the `guestbook` image stream.



8. Click the **History** menu. If you only see one entry listed here, it means OpenShift hasn’t imported your new image yet. Wait a few minutes and refresh the page. Eventually you should see a second entry, indicating that a new version of this image stream tag has been imported. This can take some time as the default frequency for importing is 15 minutes.



9. Return to the **Developer** perspective.



Note: Please wait for some time for the OpenShift console & the Developer perspective to load.

10. View the guestbook in the browser again. If you still have the tab open, go there. If not, click the Route again from the `guestbook` Deployment. You should see your new title on this page! OpenShift imported the new version of our image, and since the Deployment points to the image stream, it began running this new version as well.

Guestbook storage

1. From the guestbook in the browser, click the `/info` link beneath the input box. This is an information endpoint for the guestbook.

Not secure | guestbook-sn-labs- labs-prod-openshift-san-a45631dc5778dc6371c67d206ba9ae5c-0000.us-east.contai

Alex's Guestbook

http://guestbook-sn-labs- .labs-prod-openshift-san-a45631dc
east.containers.appdomain.cloud/
/env /info

Notice that it says "In-memory datastore (not redis)". Currently, we have only deployed the guestbook web front end, so it is using in-memory datastore to keep track of the entries. This is not very resilient, however, because any update or even a restart of the Pod will cause the entries to be lost. But let's confirm this.

Not secure | guestbook-sn-labs- .labs-prod-openshift-san-a45631dc5778dc6371c67d206ba9ae5c-0000

```
In-memory datastore (not redis)
```

2. Return to the guestbook application in the browser by clicking the Route location again. You should see that your previous entries appear no more. This is because the guestbook was restarted when your update was deployed in the last section. We need a way to persist the guestbook entries even after restarts.

Not secure | guestbook-sn-labs- .labs-prod-openshift-san-a45631dc5778dc6371c67d206ba9ae5c-0000.us-east.co

Alex's Guestbook

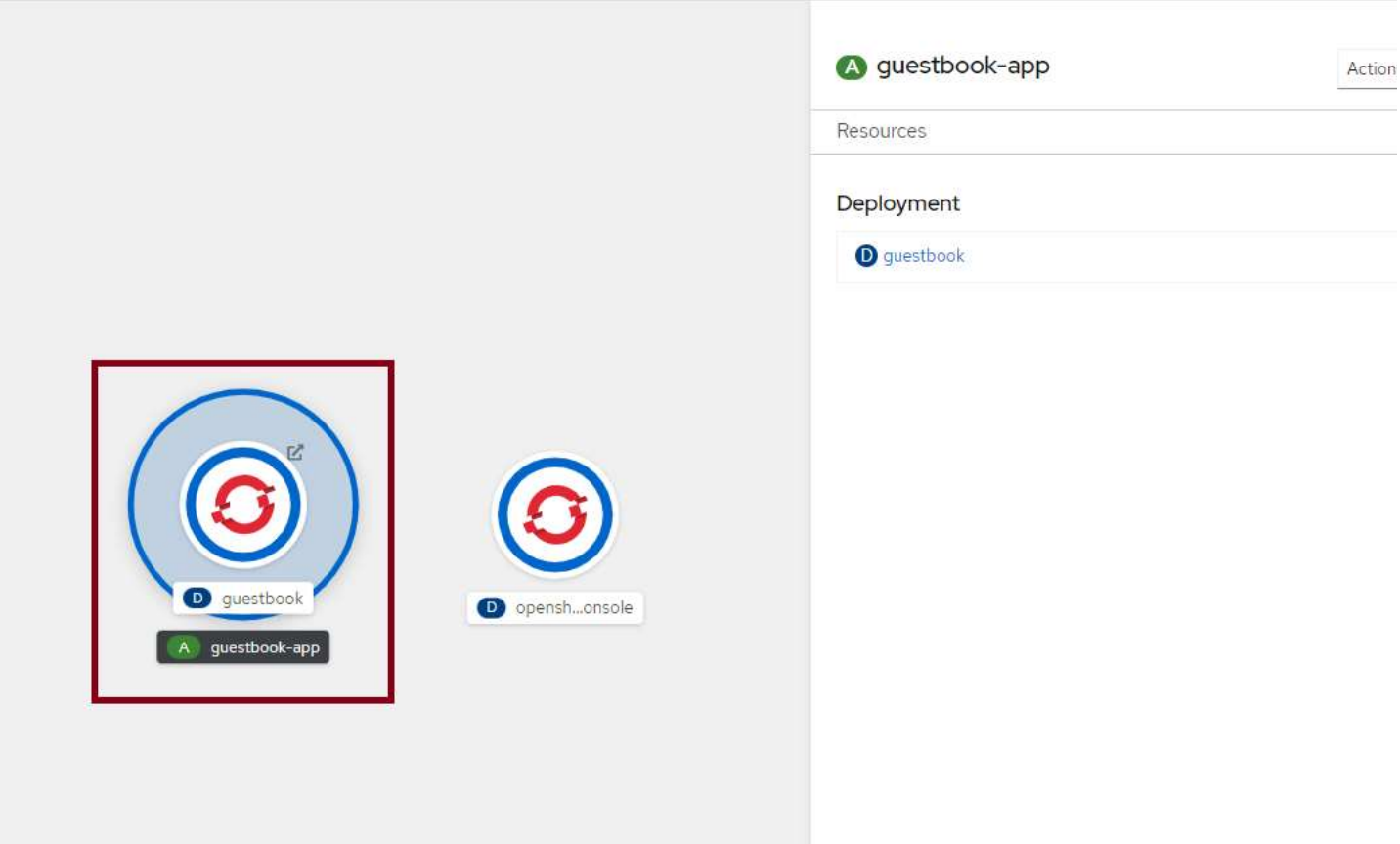
http://guestbook-sn-labs- .labs-prod-openshift-san-a45631
east.containers.appdomain.clou

Note: Currently we are experiencing certain difficulties with the OpenShift console. There is a possibility that you will see your old entries because the image stream takes time in updating. You may move ahead with the further steps of lab.

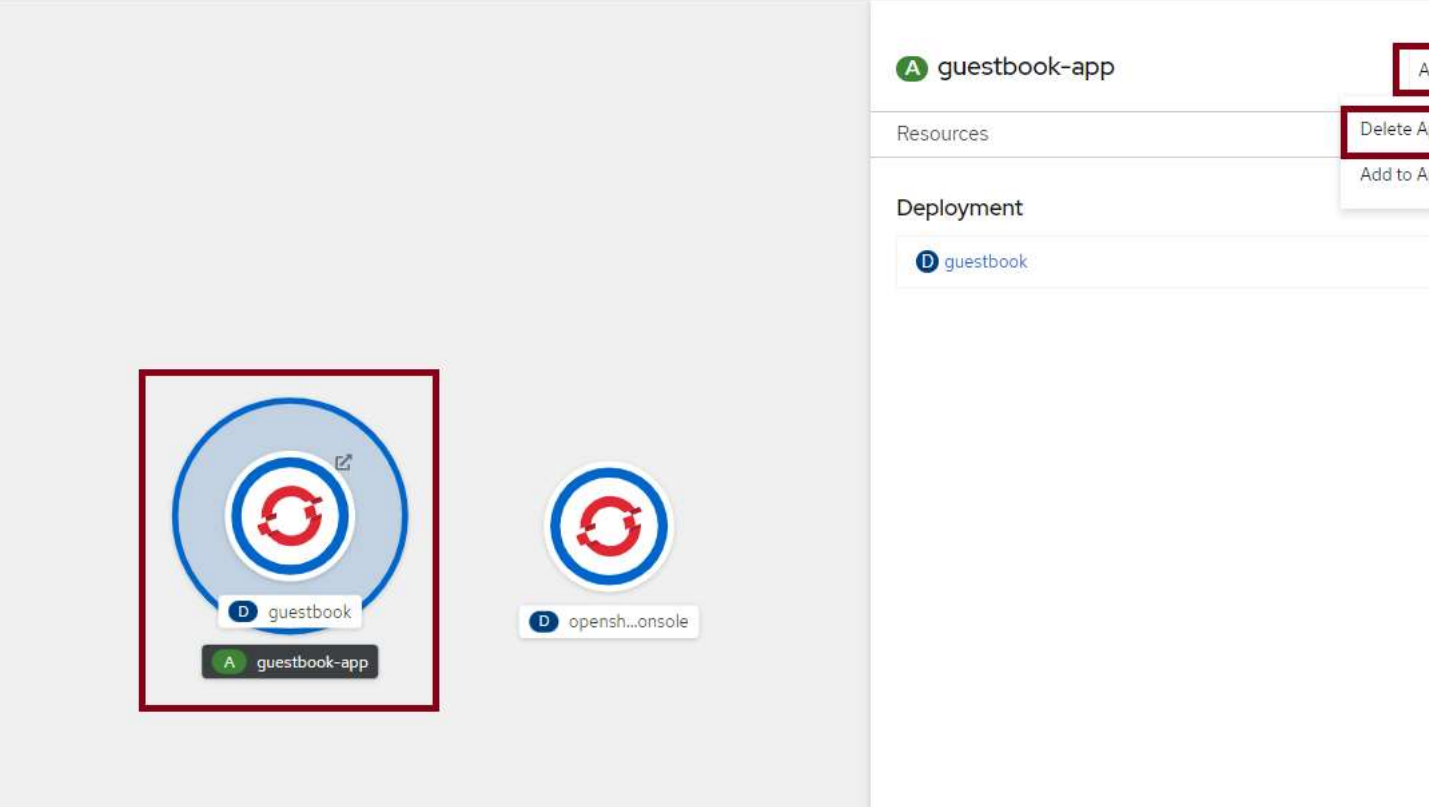
Delete the guestbook

In order to deploy a more complex version of the guestbook, delete this simple version.

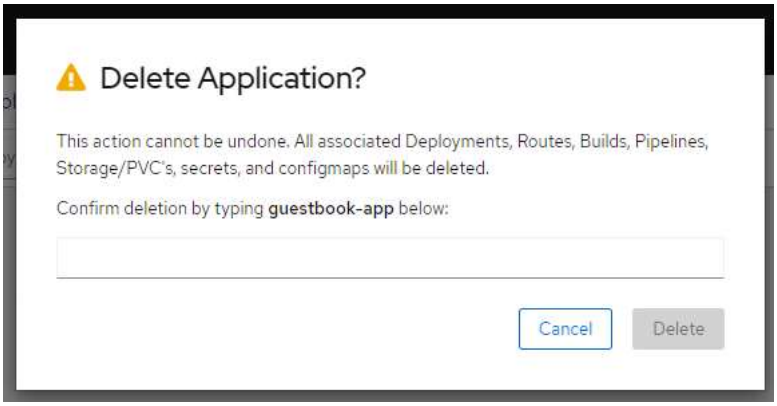
1. From the Topology view, click the `guestbook-app` application. This is the light gray circle that surrounds the `guestbook` Deployment.



2. Click **Actions** > **Delete Application**.



3. Type in the application name and click **Delete**.



Deploy Redis master and slave

We’ ve demonstrated that we need persistent storage in order for the guestbook to be effective. Let’ s deploy Redis so that we get just that. Redis is an open source, in-memory data structure store, used as a database, cache and message broker.

This application uses the v2 version of the guestbook web front end and adds on 1) a Redis master for storage and 2) a replicated set of Redis slaves. For all of these components, there are Kubernetes Deployments, Pods, and Services. One of the main concerns with building a multi-tier application on Kubernetes is resolving dependencies between all of these separately deployed components.

In a multi-tier application, there are two primary ways that service dependencies can be resolved. The `v2/guestbook/main.go` code provides examples of each. For Redis, the master endpoint is discovered through environment variables. These environment variables are set when the Redis services are started, so the service resources need to be created before the guestbook Pods start. Consequently, we’ ll follow a specific order when creating the application components. First, the Redis components will be created, then the guestbook application.

> **Note:** If you have tried this lab earlier, there might be a possibility that the previous session is still persistent. In such a case, you will see an ‘**Unchanged**’ message instead of the ‘**Created**’ message when you run the **Apply** command for creating deployments. We recommend you to proceed with the next steps of the lab.

1. From the terminal in the lab environment, change to the v2 directory.

```
1
cd ../../v2
```

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2. Run the following command or open the `redis-master-deployment.yaml` in the Explorer to familiarize yourself with the Deployment configuration for the Redis master.

```
1
cat redis-master-deployment.yaml
```

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```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ cat redis-master-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-master
  labels:
    app: redis
    role: master
spec:
  replicas: 1
  selector:
    matchLabels:
      app: redis
      role: master
  template:
    metadata:
      labels:
        app: redis
        role: master
    spec:
      containers:
      - name: redis-master
        image: redis:5.0.5
        ports:
        - name: redis-server
          containerPort: 6379
        volumeMounts:
        - name: redis-storage
          mountPath: /data
      volumes:
      - name: redis-storage
        emptyDir: {}
theia@theiaopenshift[redacted]/home/project/guestbook/v2$
```

3. Create the Redis master Deployment.

```
1
oc apply -f redis-master-deployment.yaml
```

Copied!

```
theia@theiaopenshift[redacted]/home/project/guestbook/v2$ oc apply -f redis-master-deployment.yaml
deployment.apps/redis-master created
theia@theiaopenshift[redacted]/home/project/guestbook/v2$
```

4. Verify that the Deployment was created.

```
1
oc get deployments
```

Copied!

```
theia@theiaopenshift[redacted]/home/project/guestbook/v2$ oc get deployments
NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
openshift-web-console   1/1     1            1           14m
redis-master            1/1     1            1           34s
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

5. List Pods to see the Pod created by the Deployment.

```
1
oc get pods
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ oc get pods
NAME                                    READY   STATUS    RESTARTS   AGE
openshift-web-console-77d78f965-6krl2   2/2     Running   0          14m
redis-master-d98597c5b-f2g54            1/1     Running   0          58s
theia@theiaopenshift[redacted]/home/project/guestbook/v2$
```

You can also return to the Topology view in the OpenShift web console and see that the Deployment has appeared there.

6. Run the following command or open the `redis-master-service.yaml` in the Explorer to familiarize yourself with the Service configuration for the Redis master.

```
1
cat redis-master-service.yaml
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ cat redis-master-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: redis-master
  labels:
    app: redis
    role: master
spec:
  ports:
  - port: 6379
    targetPort: redis-server
  selector:
    app: redis
    role: master
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

Services find the Pods to load balance based on Pod labels. The Pod that you created in previous step has the labels `app=redis` and `role=master`. The selector field of the Service determines which Pods will receive the traffic sent to the Service.

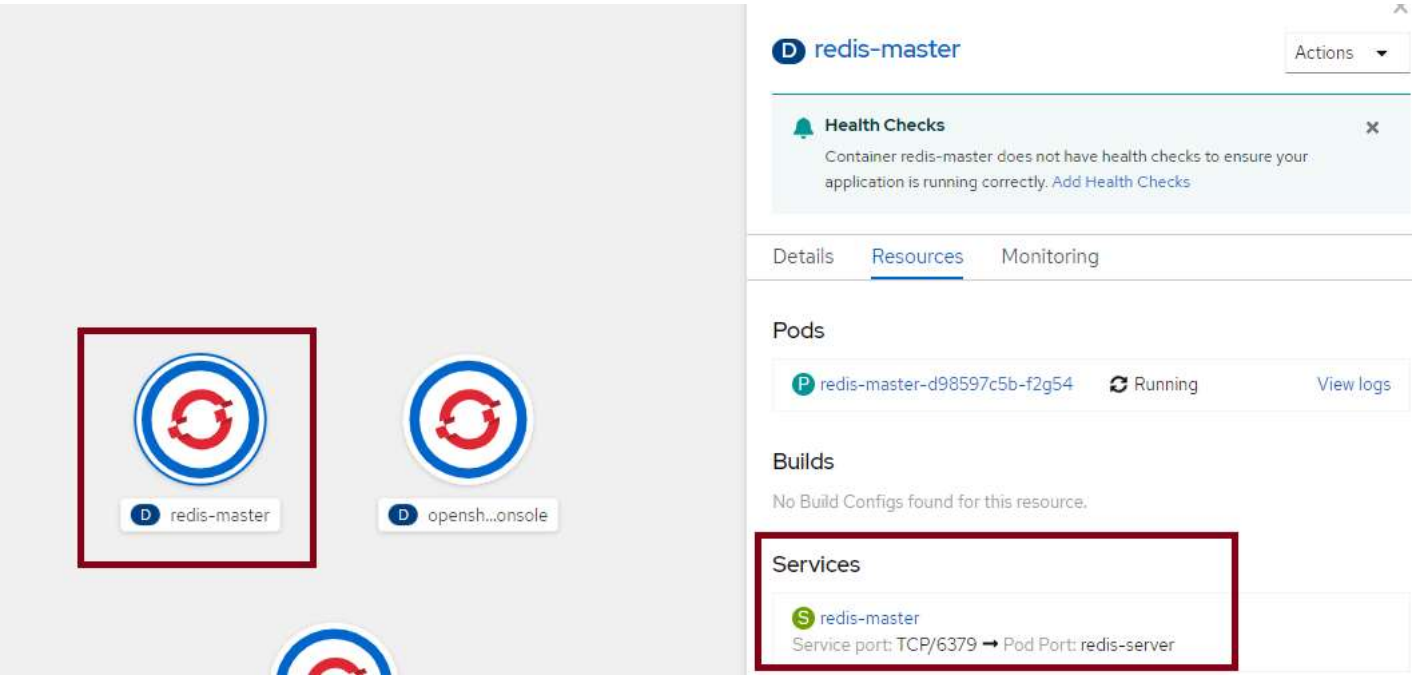
7. Create the Redis master Service.

```
1. 1
1. oc apply -f redis-master-service.yaml
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ oc apply -f redis-master-service.yaml
service/redis-master created
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

If you click on the `redis-master` Deployment in the Topology view, you should now see the `redis-master` Service in the **Resources** tab.



8. Run the following command or open the `redis-slave-deployment.yaml` in the Explorer to familiarize yourself with the Deployment configuration for the Redis slave.

```
1. 1
1. cat redis-slave-deployment.yaml
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ cat redis-slave-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-slave
  labels:
    app: redis
    role: slave
spec:
  replicas: 2
  selector:
    matchLabels:
      app: redis
      role: slave
  template:
    metadata:
      labels:
        app: redis
        role: slave
    spec:
      containers:
      - name: redis-slave
        image: redis:5.0.5
        command: ["/bin/sh"]
        args: ["-c","redis-server --slaveof redis-master 6379"]
        ports:
        - name: redis-server
          containerPort: 6379
        volumeMounts:
        - name: redis-storage
          mountPath: /data
      volumes:
      - name: redis-storage
        emptyDir: {}
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

9. Create the Redis slave Deployment.

```
1. 1
1. oc apply -f redis-slave-deployment.yaml
```

Copied!

```
theia@theiaopenshift-[redacted]:/home/project/guestbook/v2$ oc apply -f redis-slave-deployment.yaml
deployment.apps/redis-slave created
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

10. Verify that the Deployment was created.

```
1. 1
1. oc get deployments
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ oc get deployments
NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
openshift-web-console   1/1     1            1           25m
redis-master            1/1     1            1           12m
redis-slave             2/2     2            2           29s
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

11. List Pods to see the Pod created by the Deployment.

```
1. 1
1. oc get pods
```

Copied!

```
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$ oc get pods
NAME                                    READY   STATUS    RESTARTS   AGE
openshift-web-console-77d78f965-6krl2   2/2     Running   0          26m
redis-master-d98597c5b-f2g54            1/1     Running   0          13m
redis-slave-76dfcf6864-7qkmq            1/1     Running   0          66s
redis-slave-76dfcf6864-lqnnq            1/1     Running   0          66s
theia@theiaopenshift-[redacted]/home/project/guestbook/v2$
```

You can also return to the Topology view in the OpenShift web console and see that the Deployment has appeared there.

12. Run the following command or open the `redis-slave-service.yaml` in the Explorer to familiarize yourself with the Service configuration for the Redis slave.

```
1. 1
1. cat redis-slave-service.yaml
```

Copied!

```
theia@theiaopenshift /home/project/guestbook/v2$ cat redis-slave-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: redis-slave
  labels:
    app: redis
    role: slave
spec:
  ports:
  - port: 6379
    targetPort: redis-server
  selector:
    app: redis
    role: slave
```

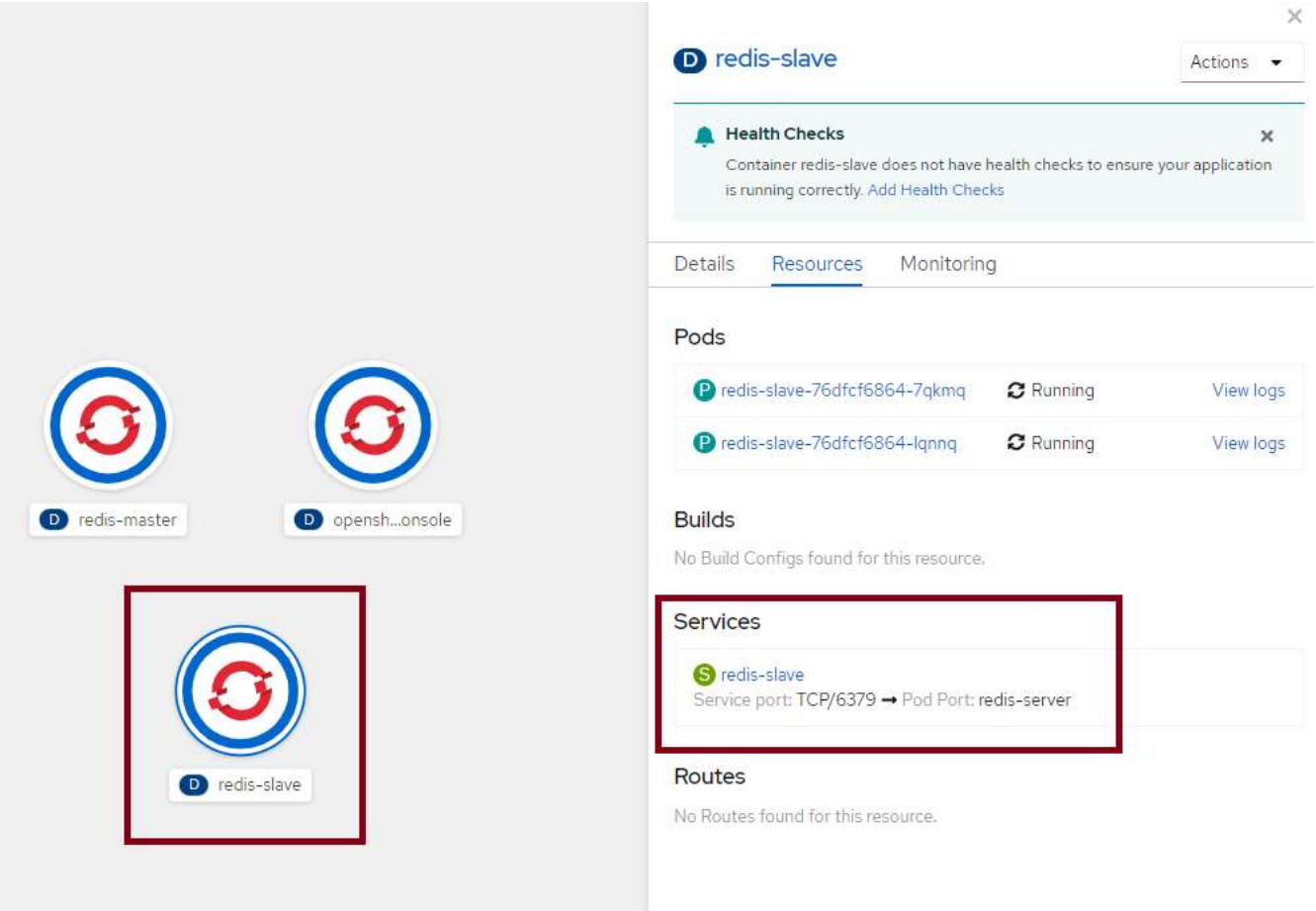
13. Create the Redis slave Service.

```
1. 1
1. oc apply -f redis-slave-service.yaml
```

Copied!

```
theia@theiaopenshift /home/project/guestbook/v2$ oc apply -f redis-slave-service.yaml
service/redis-slave created
```

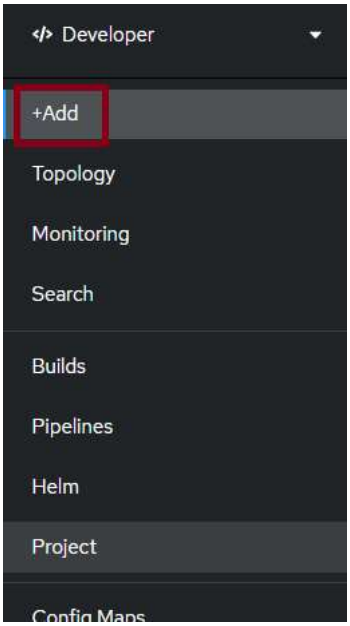
If you click on the `redis-slave` Deployment in the Topology view, you should now see the `redis-slave` Service in the **Resources** tab.



Deploy v2 guestbook app

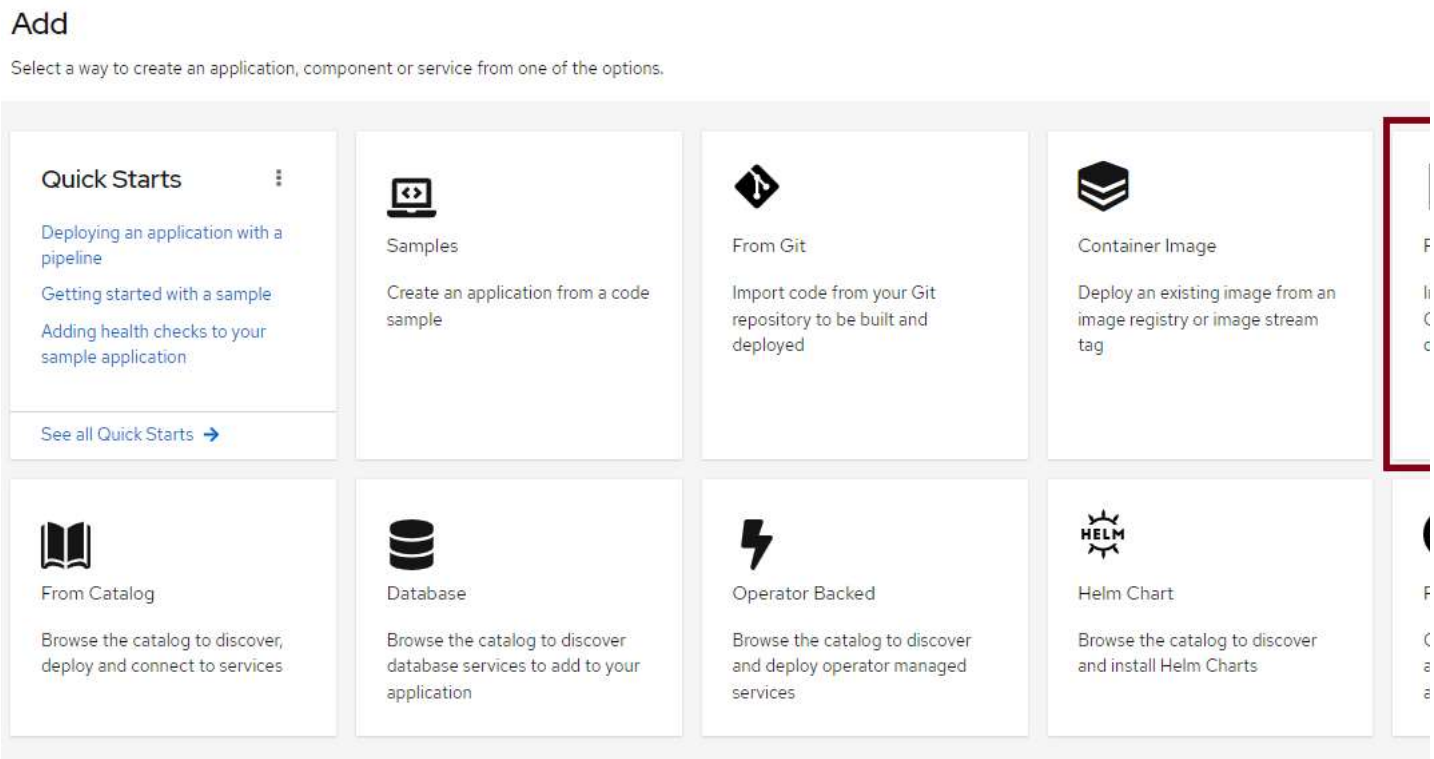
Now it's time to deploy the second version of the guestbook app, which will leverage Redis for persistent storage.

1. Click the **+Add** button to add a new application to this project.



To demonstrate the various options available in OpenShift, we' ll deploy this guestbook app using an OpenShift build and the Dockerfile from the repo.

2. Click the **From Dockerfile** option.



3. Paste the below URL in the **Git Repo URL** box.

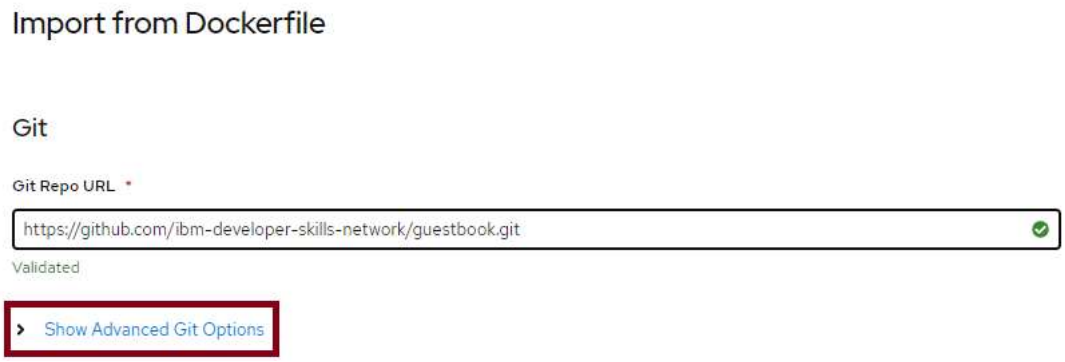
```
1. 1
1. https://github.com/ibm-developer-skills-network/guestbook
```

Copied!

You should see a validated checkmark once you click out of the box.

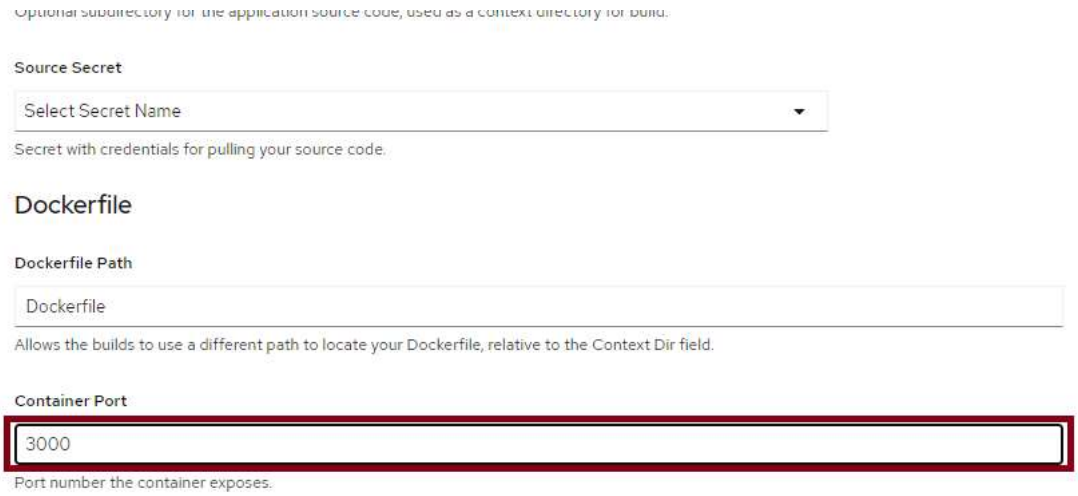
Note: Ensure there are no spaces in the Git Repo URL that is to be copied.

4. Click **Show Advanced Git Options**.

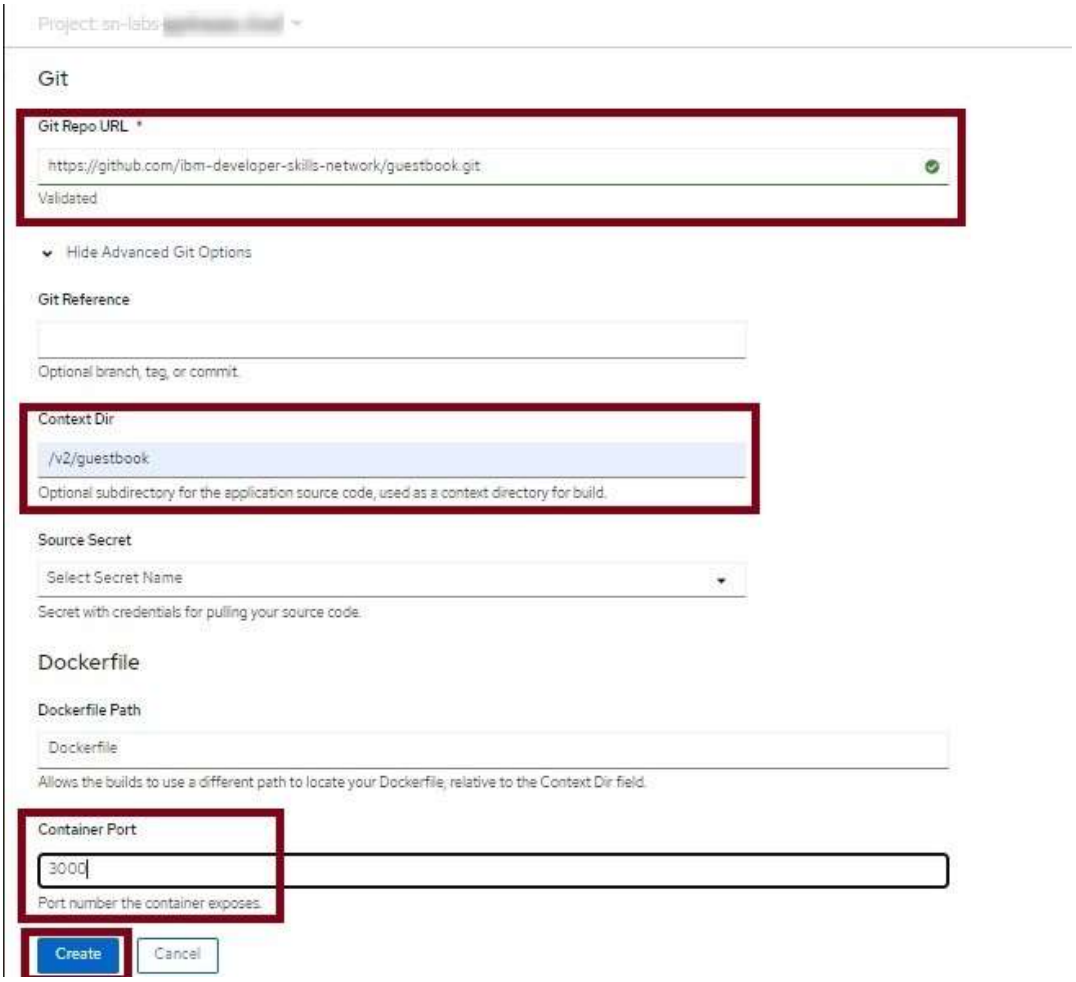


5. Since the Dockerfile isn't at the root of the repository, we need to tell OpenShift where it is. Enter `/v2/guestbook` in the **Context Dir** box.

6. Under **Container Port**, enter 3000.



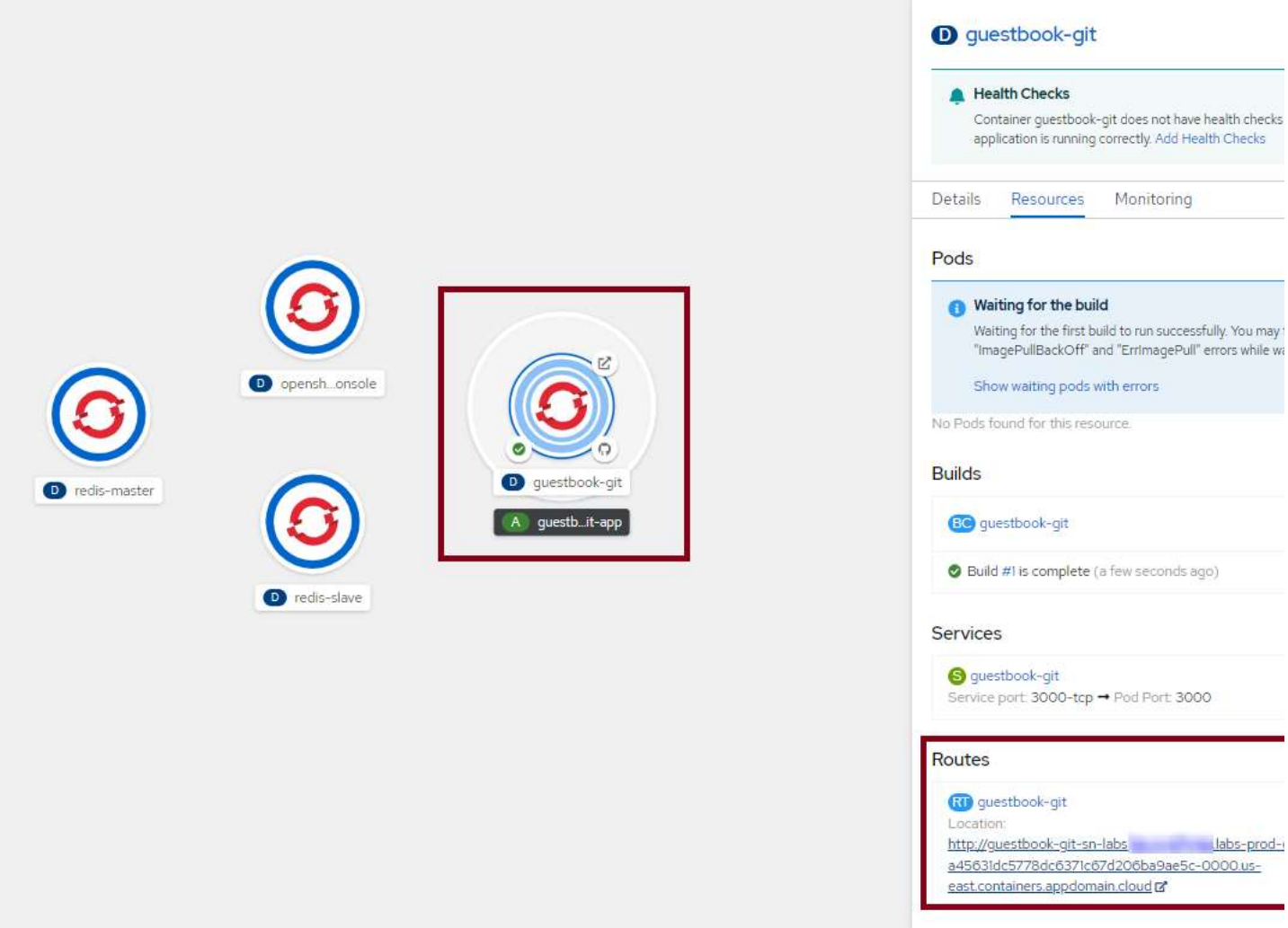
7. Leave the rest of the default options and click **Create**.
   Since we gave OpenShift a Dockerfile, it will create a BuildConfig and a Build that will build an image using the Dockerfile, push it to the internal registry, and use that image for a Deployment.



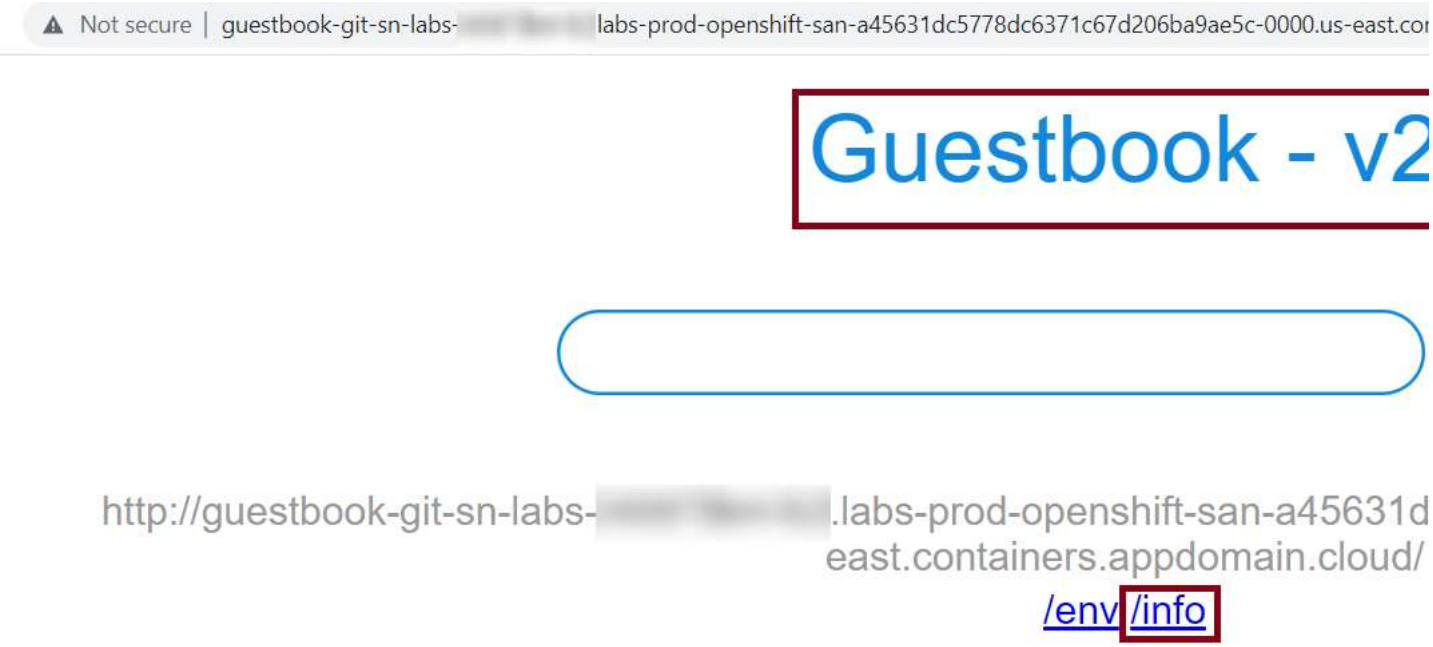
8. From the Topology view, click the `guestbook` Deployment.

In the **Resources** tab, click the Route location to load the guestbook in the browser. Notice that the header says "Guestbook - v2" instead of "Guestbook - v1".

Note: Please wait for the Builds to complete before clicking on the route link



9. From the guestbook in the browser, click the `/info` link beneath the input box.



Notice that it now gives information on Redis since we're no longer using the in-memory datastore.

Not secure | guestbook-git-sn-labs-[redacted].labs-prod-openshift-san-a45631dc5778dc6371c67d206ba9ae5c-(

```
# Server
redis_version:5.0.5
redis_git_sha1:00000000
redis_git_dirty:0
redis_build_id:442b43d467cd2b03
redis_mode:standalone
os:Linux 3.10.0-1160.59.1.el7.x86_64 x86_64
arch_bits:64
multiplexing_api:epoll
atomicvar_api:atomic-builtin
gcc_version:8.3.0
process_id:1
run_id:ddfb01eefa82ace731a16c65dae1ddc6cbe031d3
tcp_port:6379
uptime_in_seconds:1637
uptime_in_days:0
hz:10
configured_hz:10
lru_clock:5500602
executable:/data/redis-server
config_file:

# Clients
connected_clients:1
client_recent_max_input_buffer:2
client_recent_max_output_buffer:0
blocked_clients:0

# Memory
used_memory:1944584
used_memory_human:1.85M
used_memory_rss:10461184
used_memory_rss_human:9.98M
used_memory_peak:1963600
used_memory_peak_human:1.87M
used_memory_peak_perc:99.03%
used_memory_overhead:1923354
used_memory_startup:791240
used_memory_dataset:21230
used_memory_dataset_perc:1.84%
allocator_allocated:1919168
allocator_active:2138112
allocator_resident:5660672
```

- If you wish to delete & redeploy your app on Openshift due to session persistence or other errors, please follow the steps given here

  After doing the above, if you do not see any route to your `guestbook` app, please run the below command in the terminal to get the app route:

```
1
oc status
```

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The guestbook app may show a 'Waiting for Database connection' status for some time after clicking on the route, due to which, the entries added in the box will not appear. You may have to wait for sometime for the app to be ready and then add your entries to see them appear correctly.

Changelog

Date	Version	Changed by	Change Description
2022-04-12	1.1	K Sundararajan	Updated Lab instructions
2022-04-13	1.2	K Sundararajan	Updated Lab instructions
2022-04-14	1.3	K Sundararajan	Updated Lab instructions
2022-04-19	1.4	K Sundararajan	Updated Lab instructions
2022-07-25	1.5	K Sundararajan	Updated Lab instructions
2022-08-02	1.6	K Sundararajan	Added new IDSN logo
2022-10-10	1.7	K Sundararajan	Added link to new OC redeployment lab with additional notes
2022-10-11	1.8	Lavanya Rajalingam	Updated Lab instructions
2023-04-19	1.9	Sneha R Baddi	Minor update for clarity

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